电子设计基础训练

电子信息工程学院
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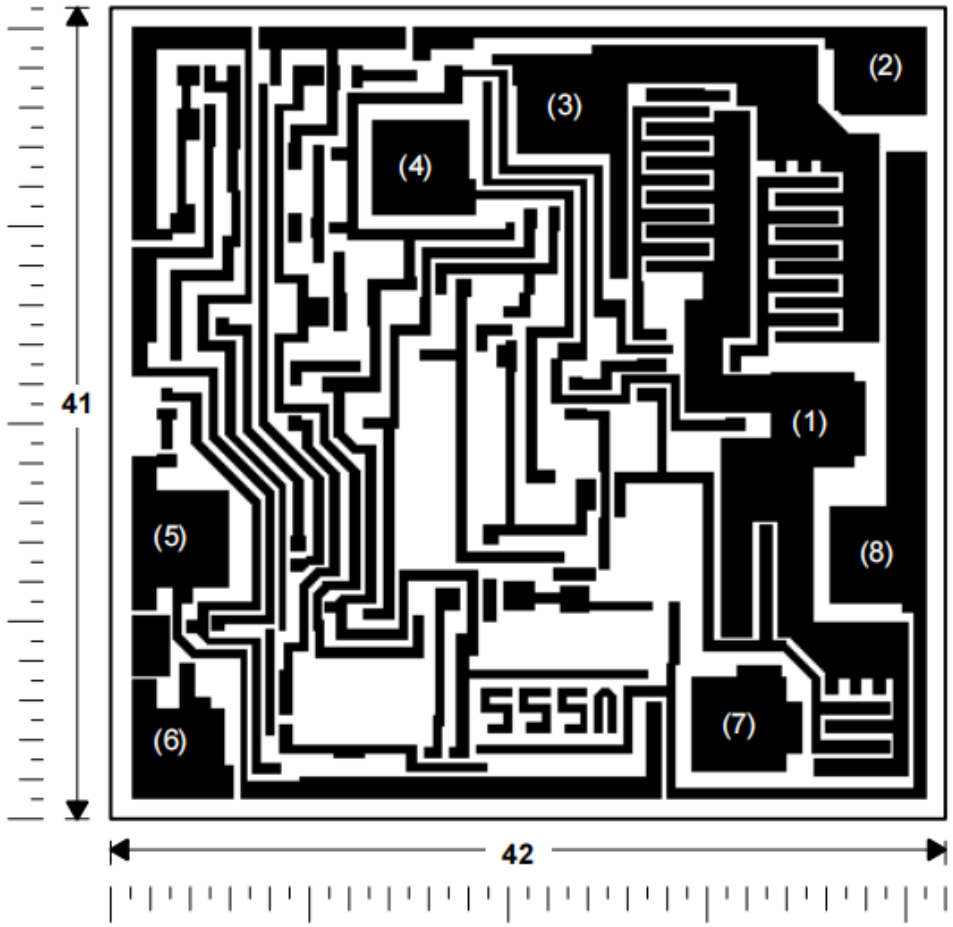
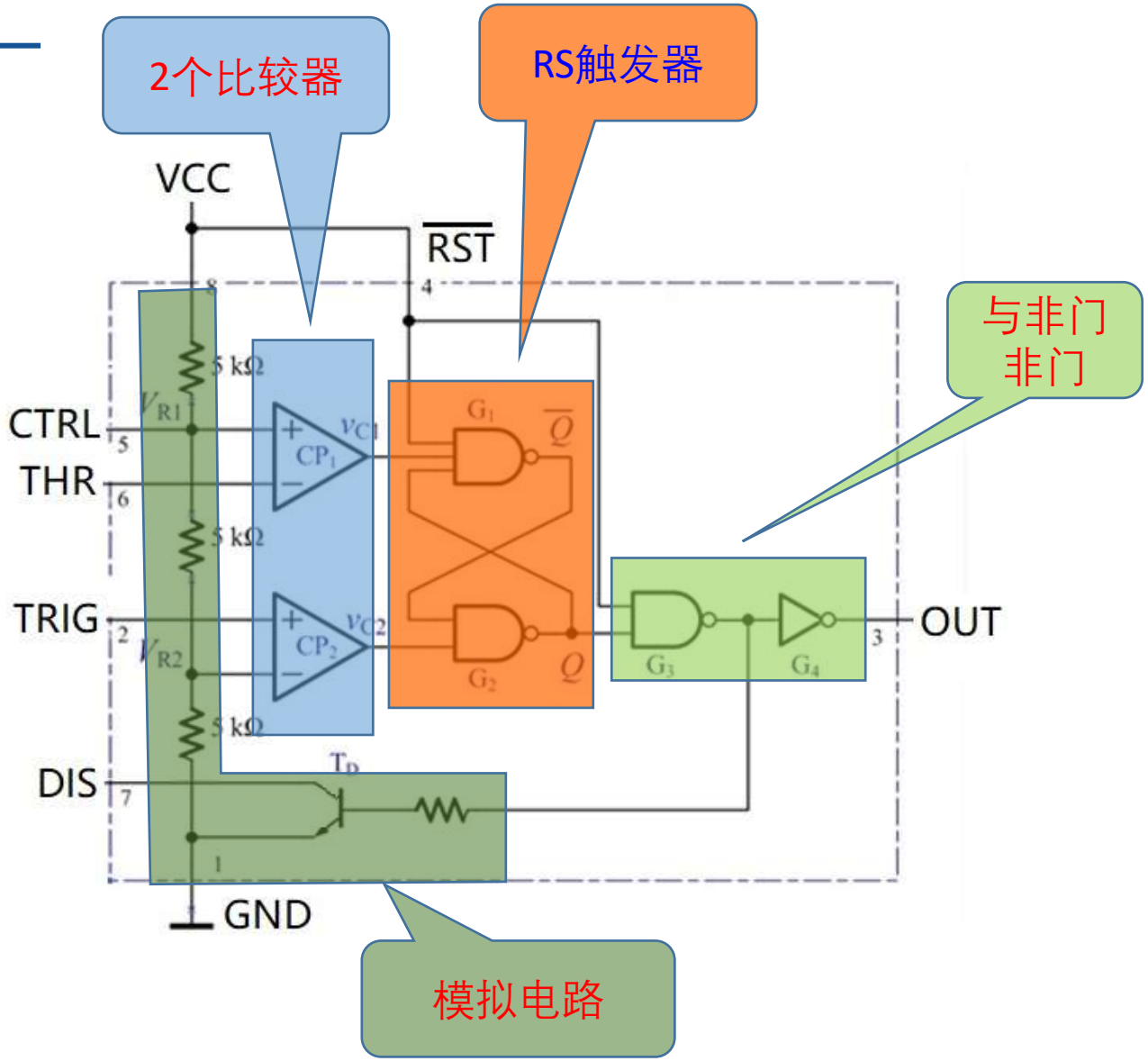
电子设计基础训练

- 逻辑概念与逻辑电平
- 逻辑运算
- 组合逻辑与时序逻辑
- 二极管、三极管搭建门电路
- 常用逻辑门芯片认识与测量
- 译码器、比较器、选择器芯片认识与测量
- 555振荡器认识与测量
- 触发器芯片认识与测量
- 计数器与移位寄存器芯片认识与测量

555振荡器

■ NE555芯片





555振荡器

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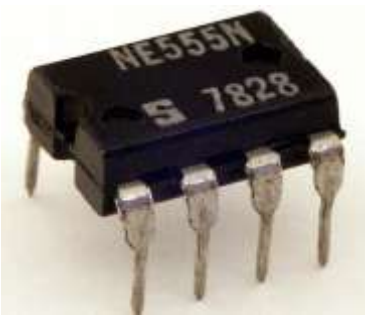
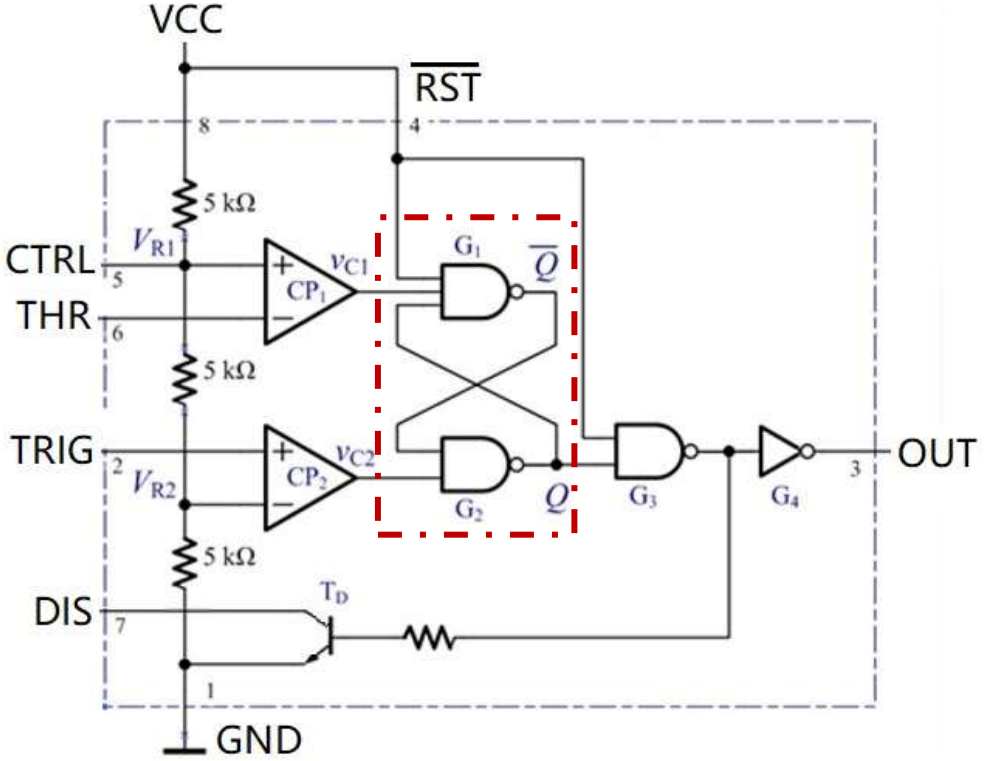


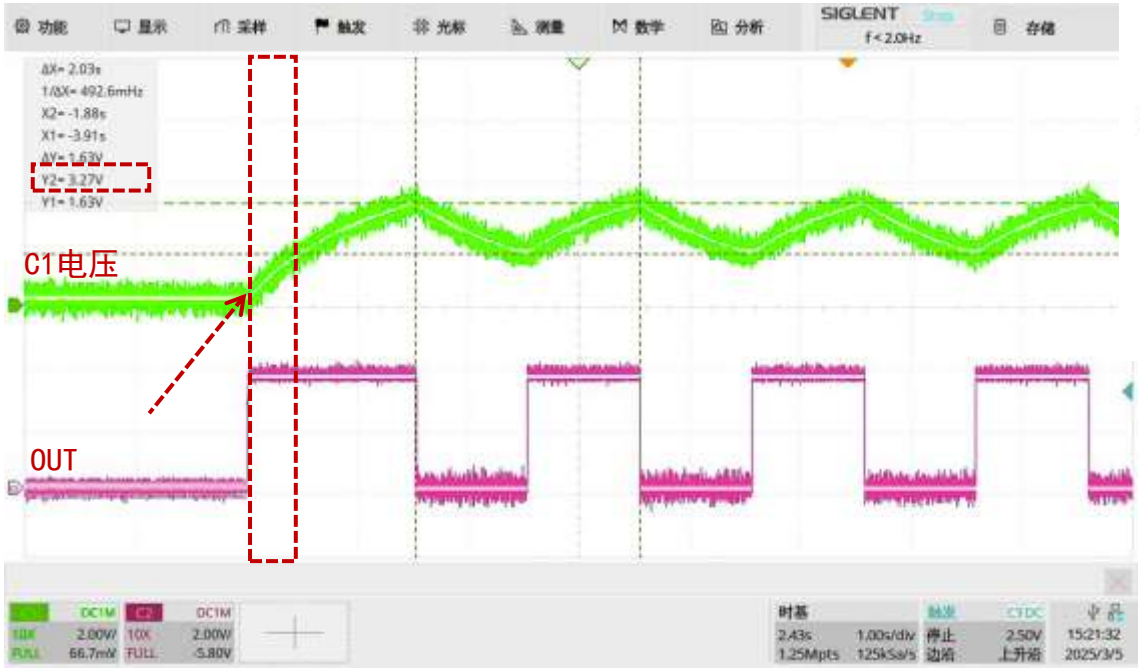
引脚	名称	功能
1	GND（地）	接地，作为低电平（0V）
2	TRIG（触发）	电压低于1/3V_{CC}时触发，输出1，并锁住
3	OUT（输出）	数字输出，与7脚（放电）逻辑一致
4	$\overline{RST}$（复位）	接地时芯片输出为0
5	CTRL（控制）	拓展实验使用，前两个实验中通过电容接地
6	THR（阈值）	电压高于2/3V_{CC}时复位，输出0
7	DIS（放电）	内接三极管，导通或截止控制外部电容放电
8	V_{CC}（供电）	提供高电平并给芯片供电

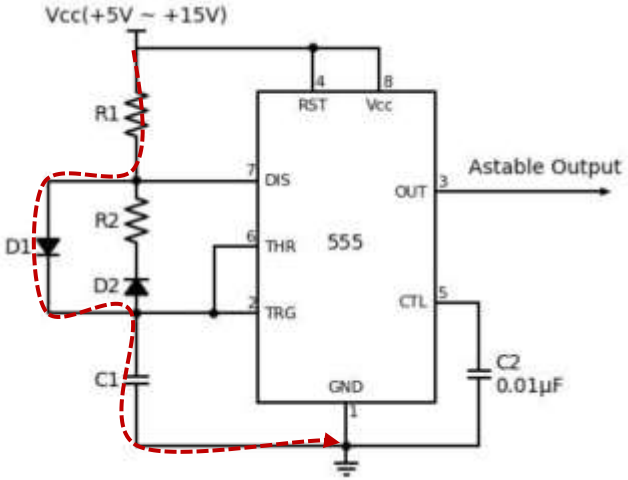


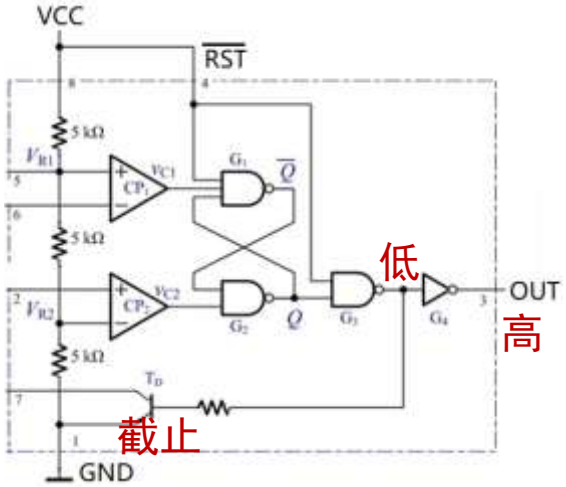
555振荡器

■ NE555多谐振荡器

➢ 初始状态下，电容C1通过VCC-R1-D1-C1-GND这一回路从0开始充电，输出为高电平





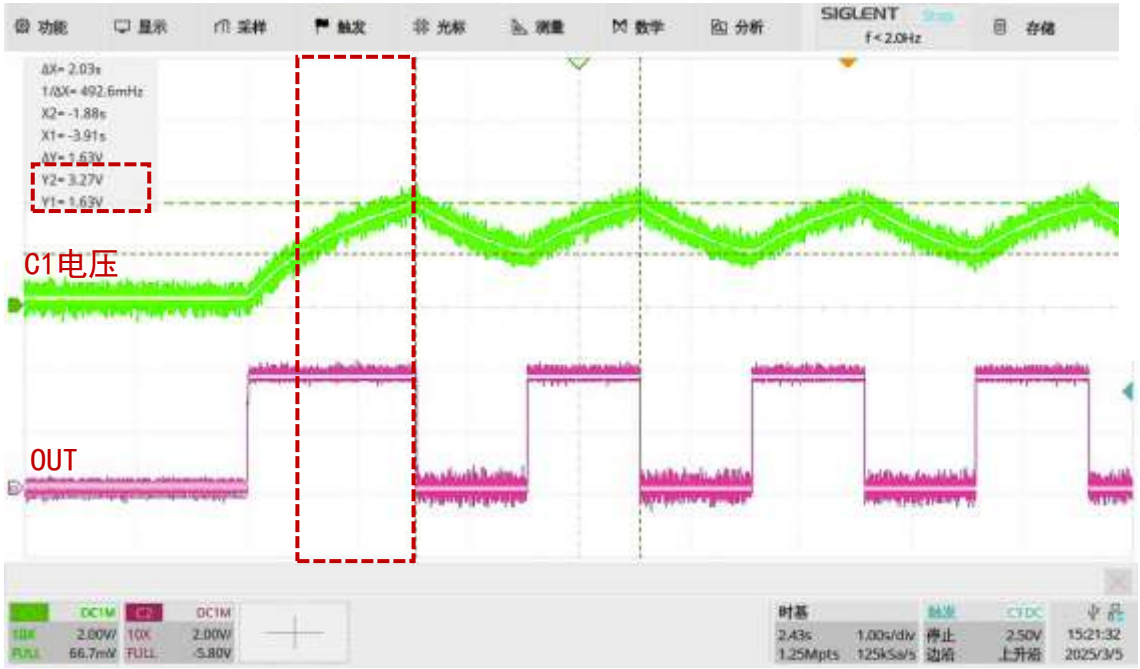


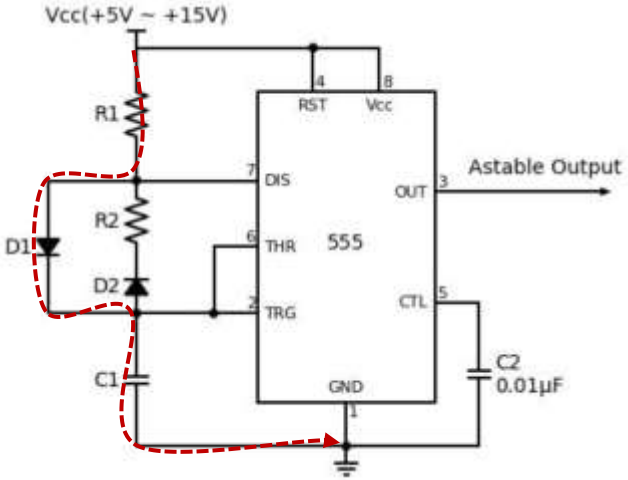
输入	输出	
2 脚 TRIG 和 6 脚 THR	3 脚 OUT	7 脚 DIS
$<\frac{1}{3}VCC$	高	高阻态
$(\frac{1}{3}VCC,\frac{2}{3}VCC)$	保持不变	保持不变
$>\frac{2}{3}VCC$	低	低

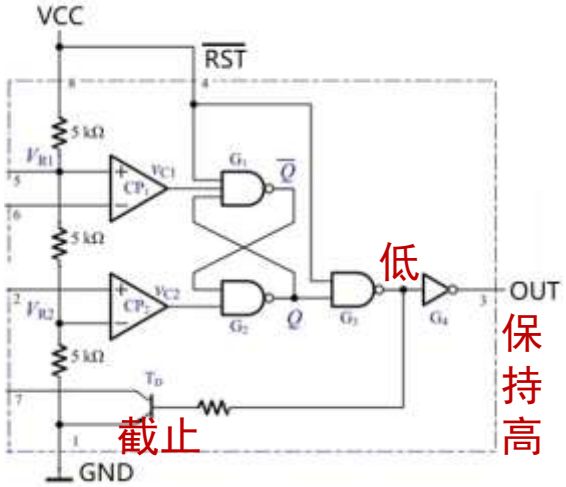
555振荡器

■ NE555多谐振荡器

➢ 电容C1通过VCC-R1-D1-C1-GND充电到1/3-2/3VCC时，输出保持高电平





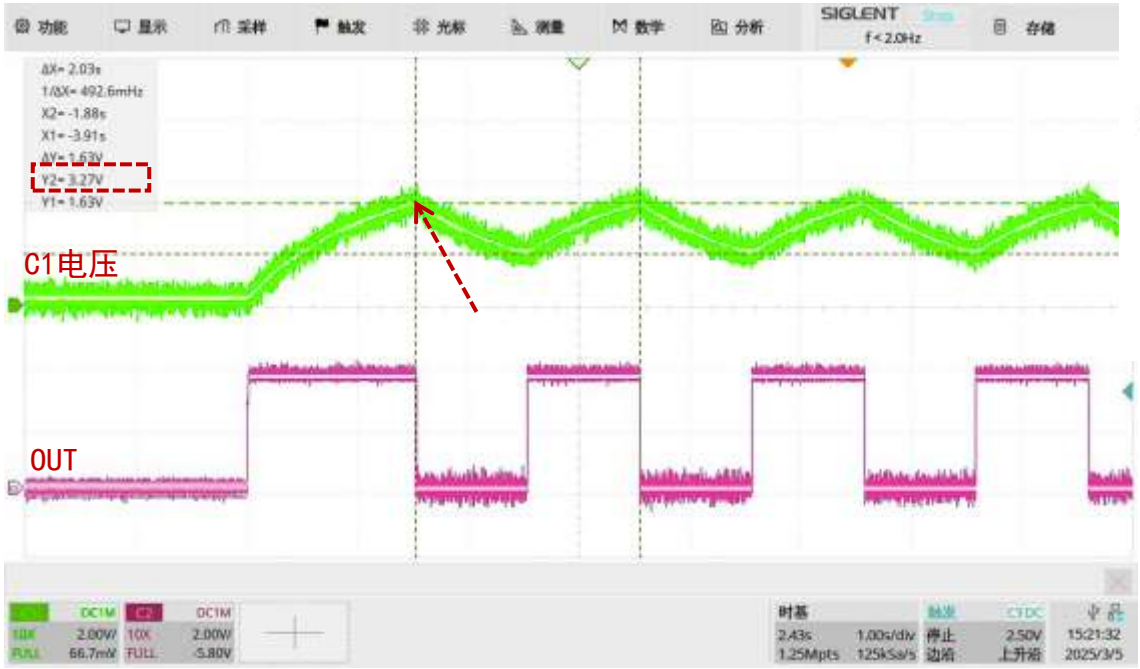


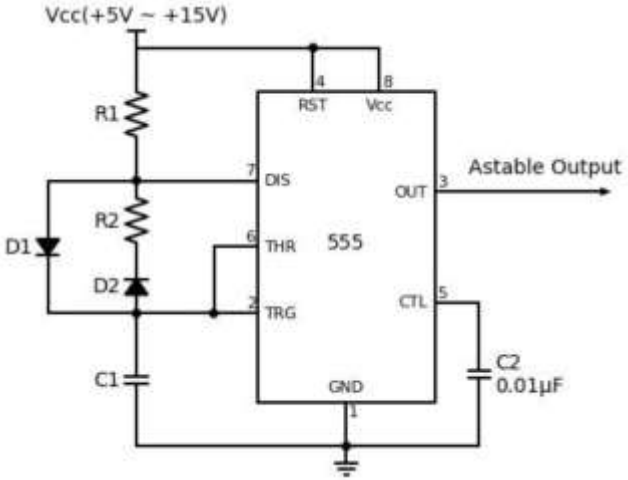
输入	输出	
2 脚 TRIG 和 6 脚 THR	3 脚 OUT	7 脚 DIS
$<\frac{1}{3}VCC$	高	高阻态
$(\frac{1}{3}VCC,\frac{2}{3}VCC)$	保持不变	保持不变
$>\frac{2}{3}VCC$	低	低

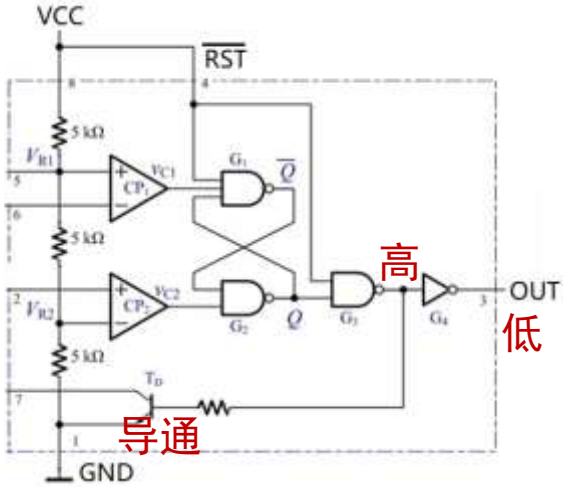
555振荡器

■ NE555多谐振荡器

➢ 电容C1通过VCC-R1-D1-C1-GND充电到2/3VCC的瞬间，输出变为低电平





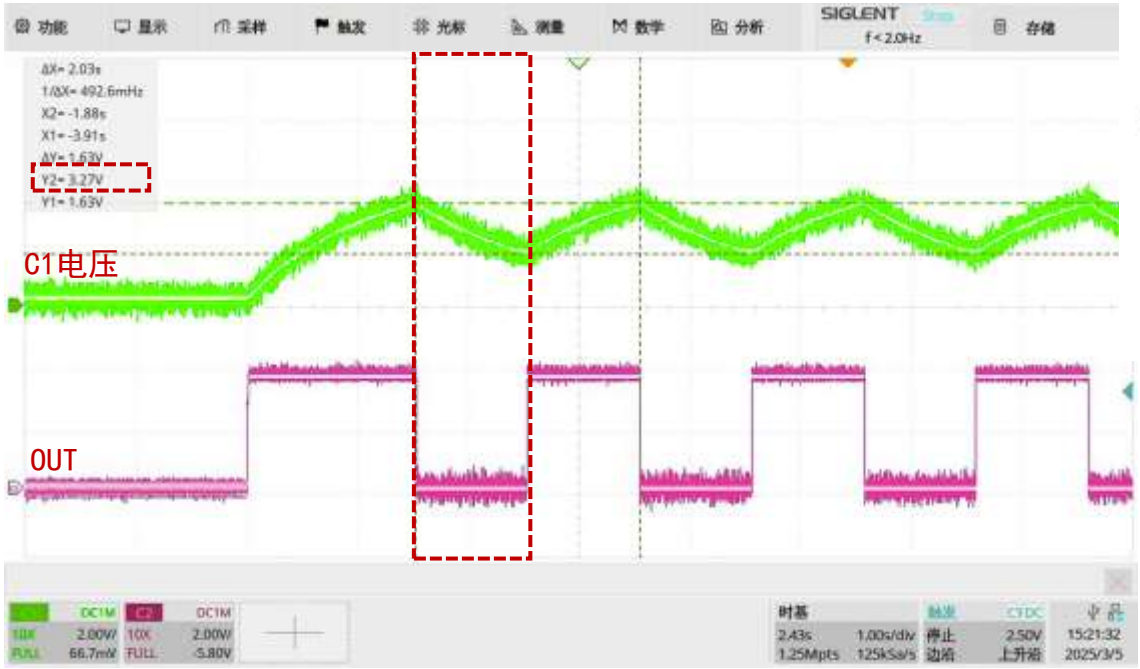


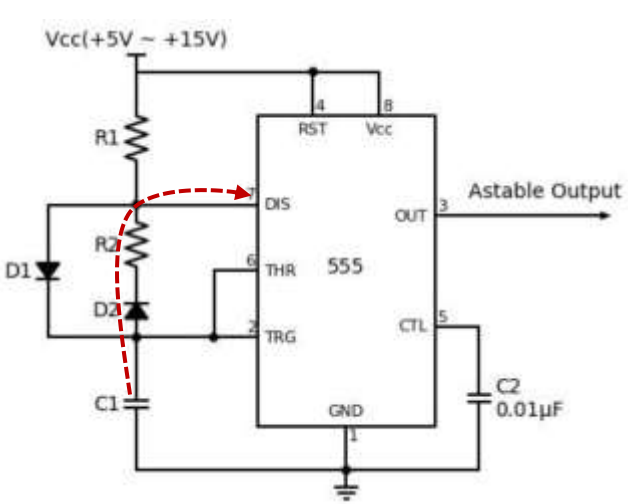
输入	输出	
2 脚 TRIG 和 6 脚 THR	3 脚 OUT	7 脚 DIS
$<\frac{1}{3}VCC$	高	高阻态
$(\frac{1}{3}VCC,\frac{2}{3}VCC)$	保持不变	保持不变
$>\frac{2}{3}VCC$	低	低

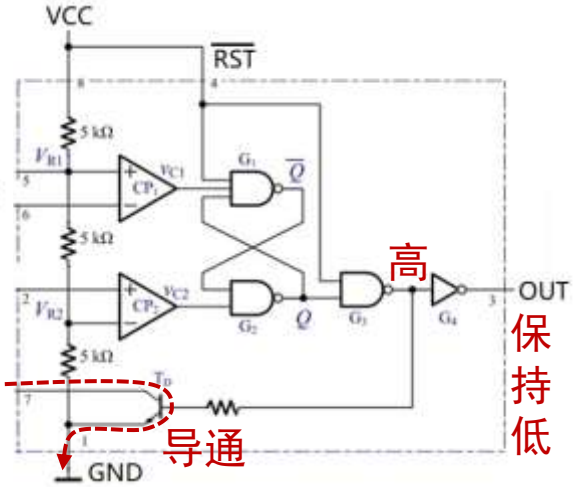
555振荡器

■ NE555多谐振荡器

➢ 电容C1通过C1-D2-R2-DIS-TD-GND放电， 当电压处于1/3-2/3VCC时当输出保持低电平





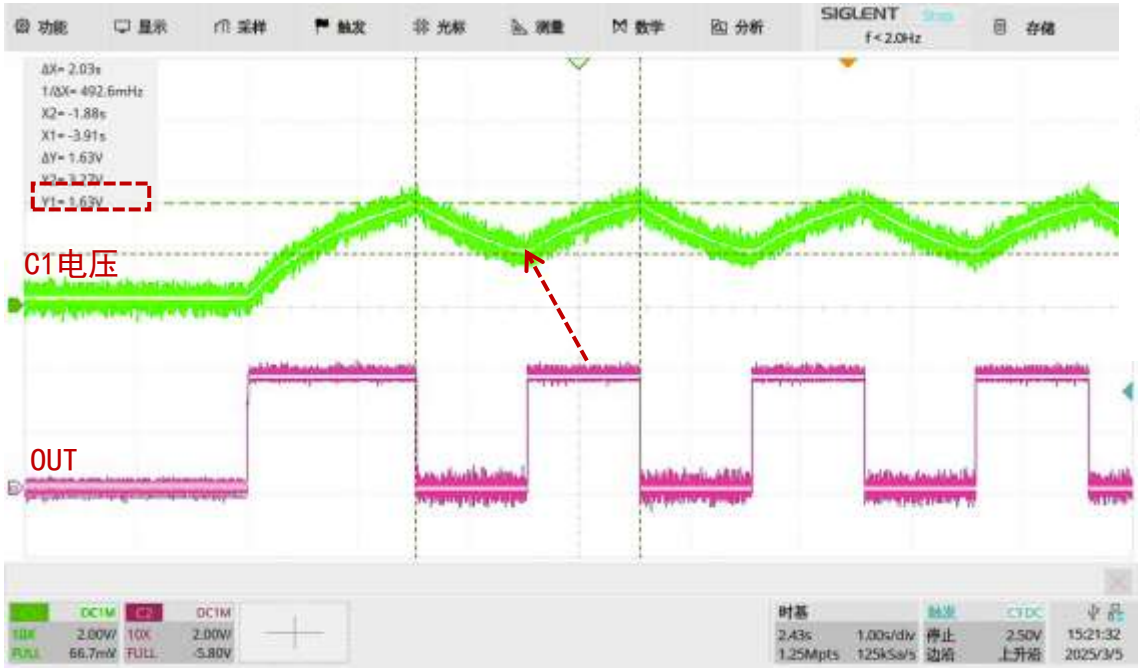


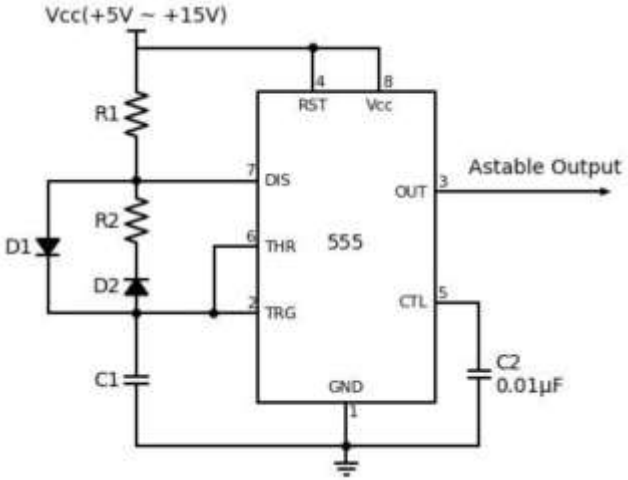
输入	输出	
2 脚 TRIG 和 6 脚 THR	3 脚 OUT	7 脚 DIS
$<\frac{1}{3}VCC$	高	高阻态
$(\frac{1}{3}VCC,\frac{2}{3}VCC)$	保持不变	保持不变
$>\frac{2}{3}VCC$	低	低

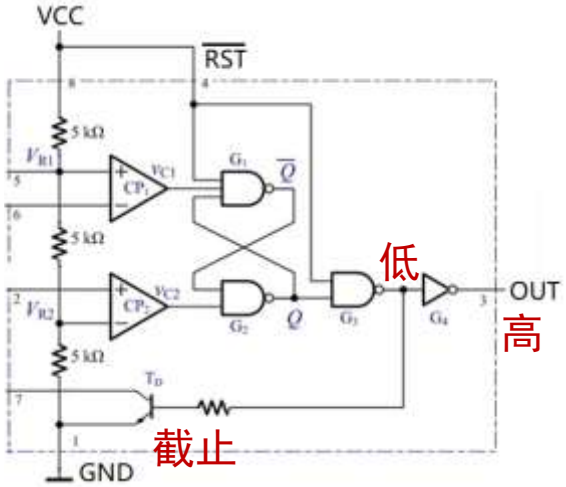
555振荡器

■ NE555多谐振荡器

➢ 当电容C1放电到电压小于1/3VCC的瞬间，输出高电平，三级管截止，停止放电，后续再次充电







输入	输出	
2 脚 TRIG 和 6 脚 THR	3 脚 OUT	7 脚 DIS
$<\frac{1}{3}VCC$	高	高阻态
$(\frac{1}{3}VCC,\frac{2}{3}VCC)$	保持不变	保持不变
$>\frac{2}{3}VCC$	低	低

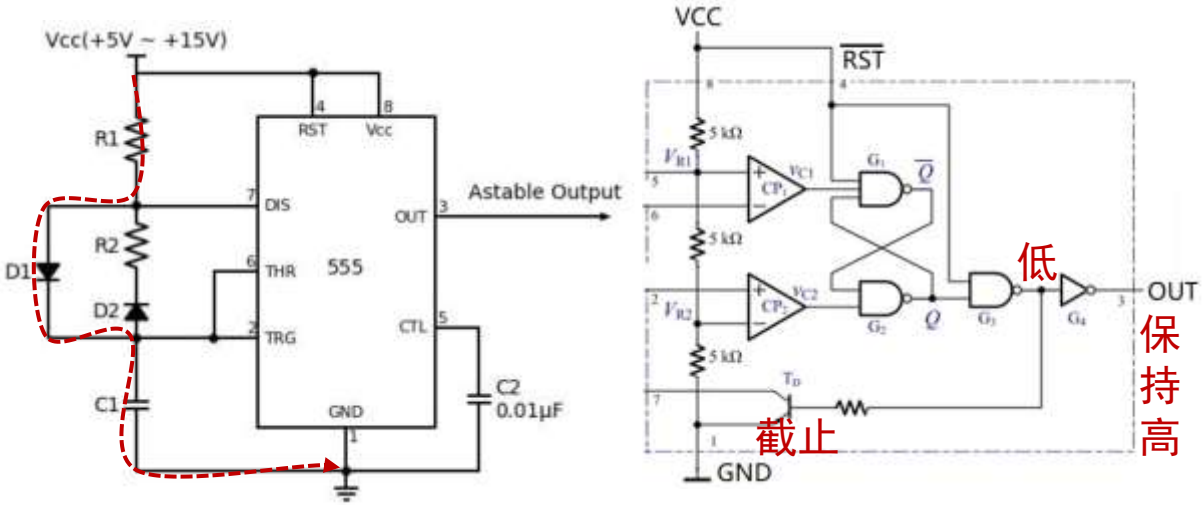
555振荡器

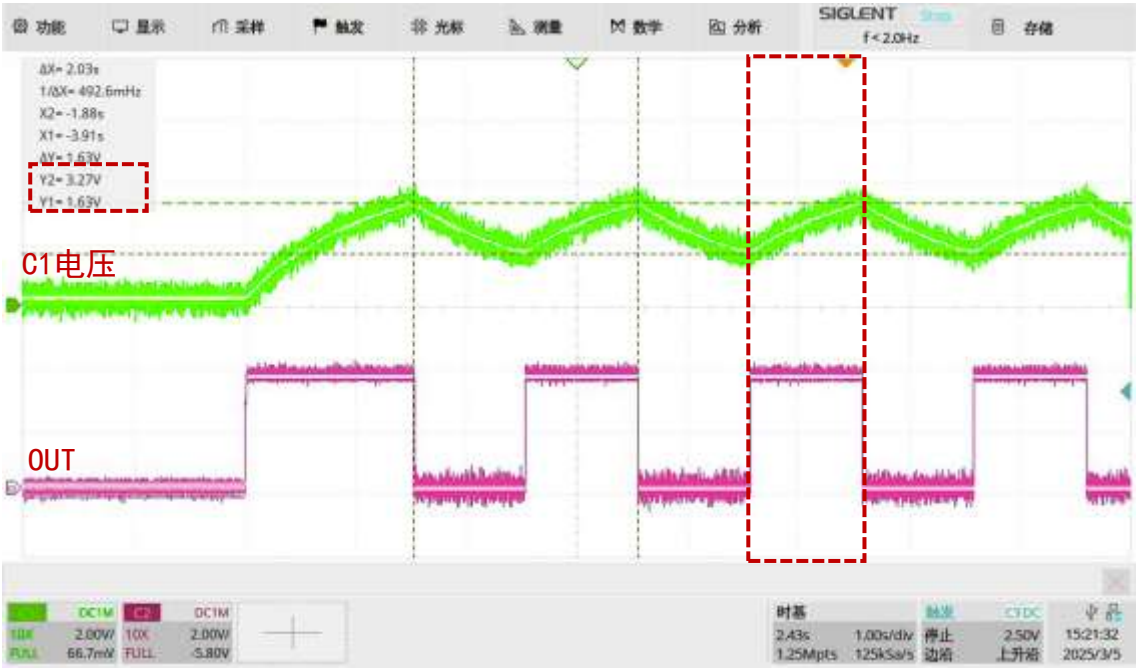
■ NE555多谐振荡器

➢ 高电平时间计算：根据电容充放电时间公式

$$\Delta t = t - t_0 = RC\ln\left(\frac{v_c(\infty) - v_c(t_0)}{v_c(\infty) - v_c(t)}\right)$$

$$\Delta t_1 = R_1C_1\ln\left(\frac{VCC - V_D - \frac{1}{3}VCC}{VCC - V_D - \frac{2}{3}VCC}\right) = R_1C_1\ln\left(\frac{V_D - \frac{2}{3}VCC}{V_D - \frac{1}{3}VCC}\right)$$





555振荡器

$$\Delta t_1 = R_1C_1\ln\left(\frac{VCC - V_D - \frac{1}{3}VCC}{VCC - V_D - \frac{2}{3}VCC}\right) = R_1C_1\ln\left(\frac{V_D - \frac{2}{3}VCC}{V_D - \frac{1}{3}VCC}\right)$$

■ NE555多谐振荡器

➢ 低电平时间计算：根据电容充放电时间公式

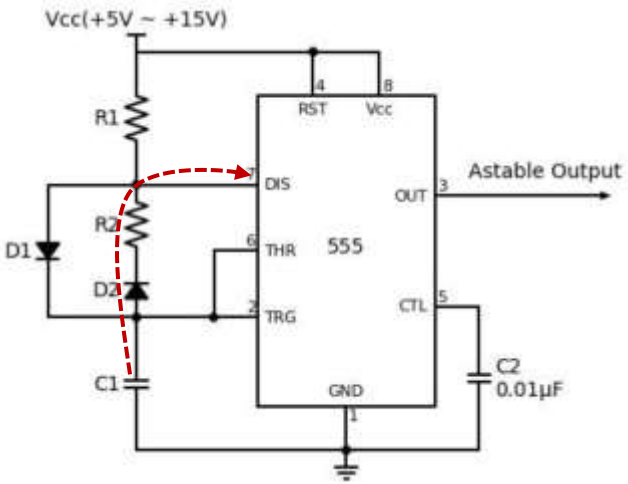
$$\Delta t = t - t_0 = RC\ln\left(\frac{v_c(\infty) - v_c(t_0)}{v_c(\infty) - v_c(t)}\right)$$

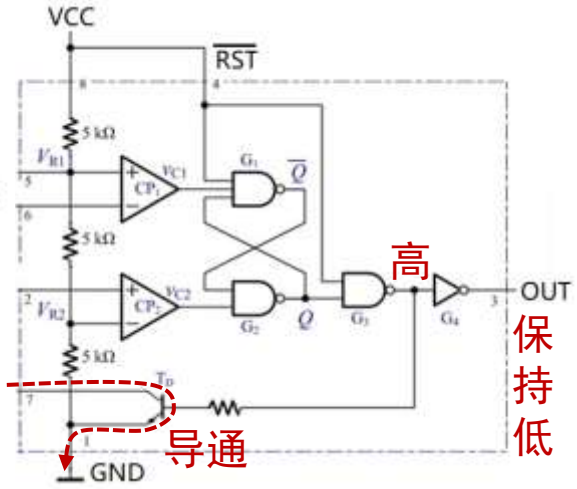
$$\Delta t_2 = R_2C_1\ln\left(\frac{V_D - \frac{2}{3}VCC}{V_D - \frac{1}{3}VCC}\right)$$

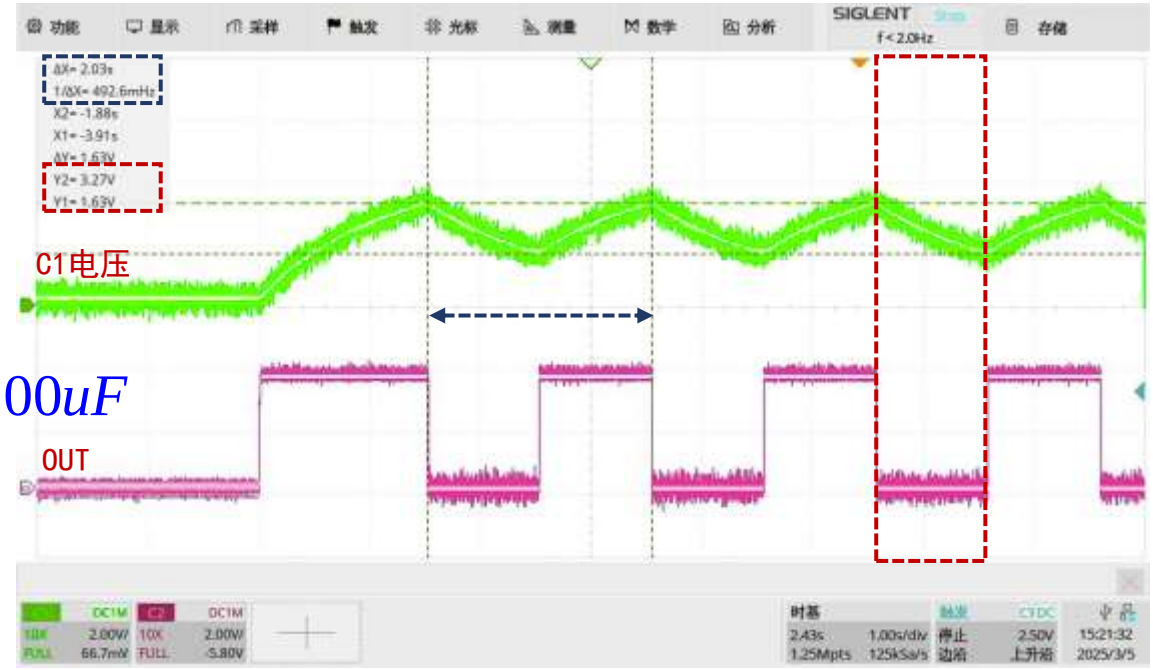
➢ $\Delta t_1 = \Delta t_2$，占空比50%；代入

$V_D = 0.7V$，$VCC = 5V$，$R_1 = 10k\Omega$，$C_1 = 100uF$

得出方波周期：$T = \Delta t_1 + \Delta t_2 = 2s$



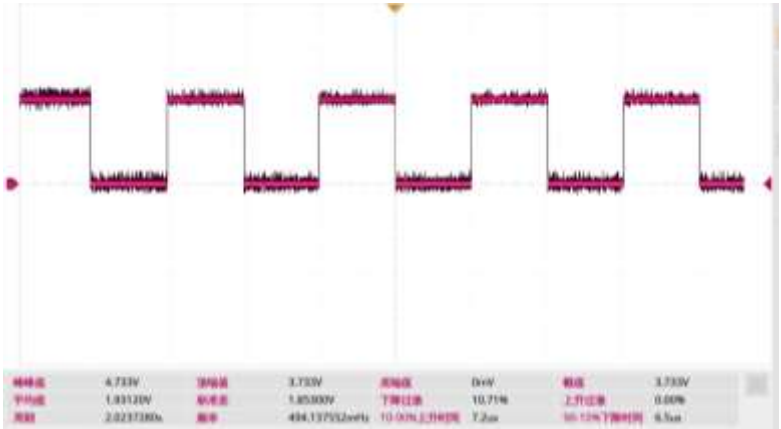


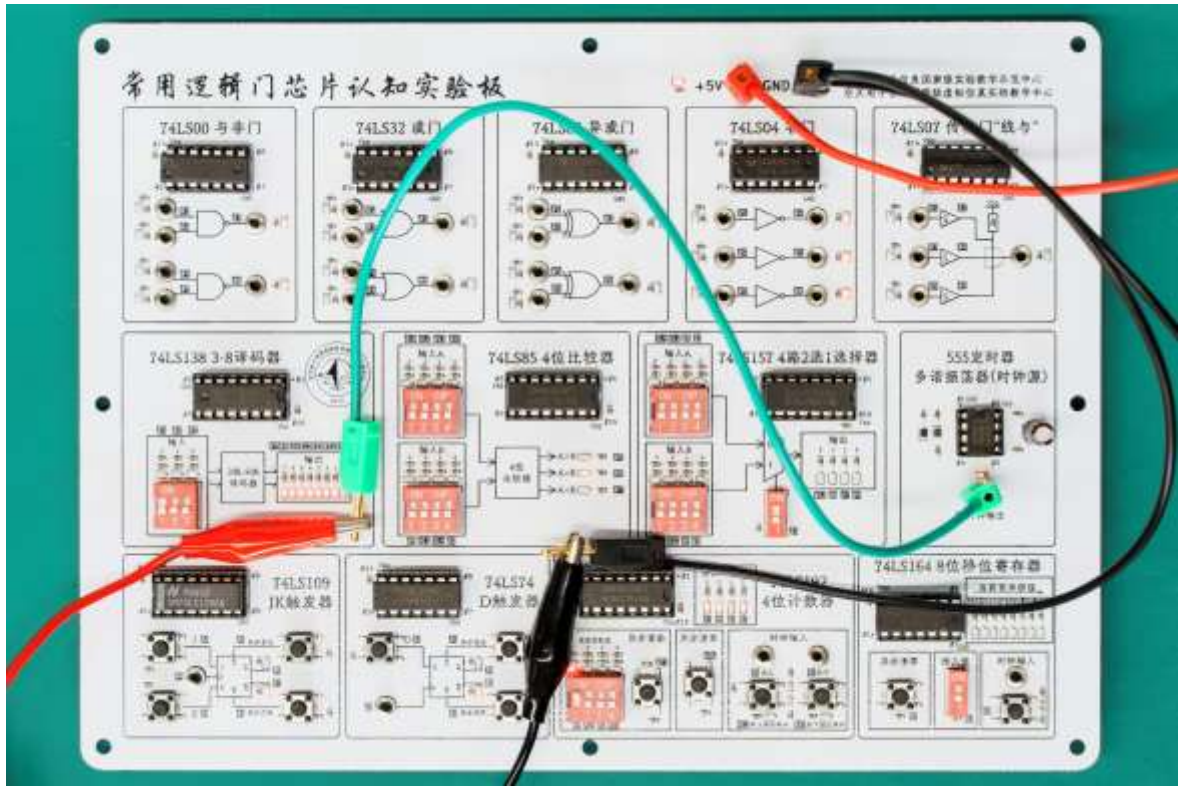


555振荡器实验

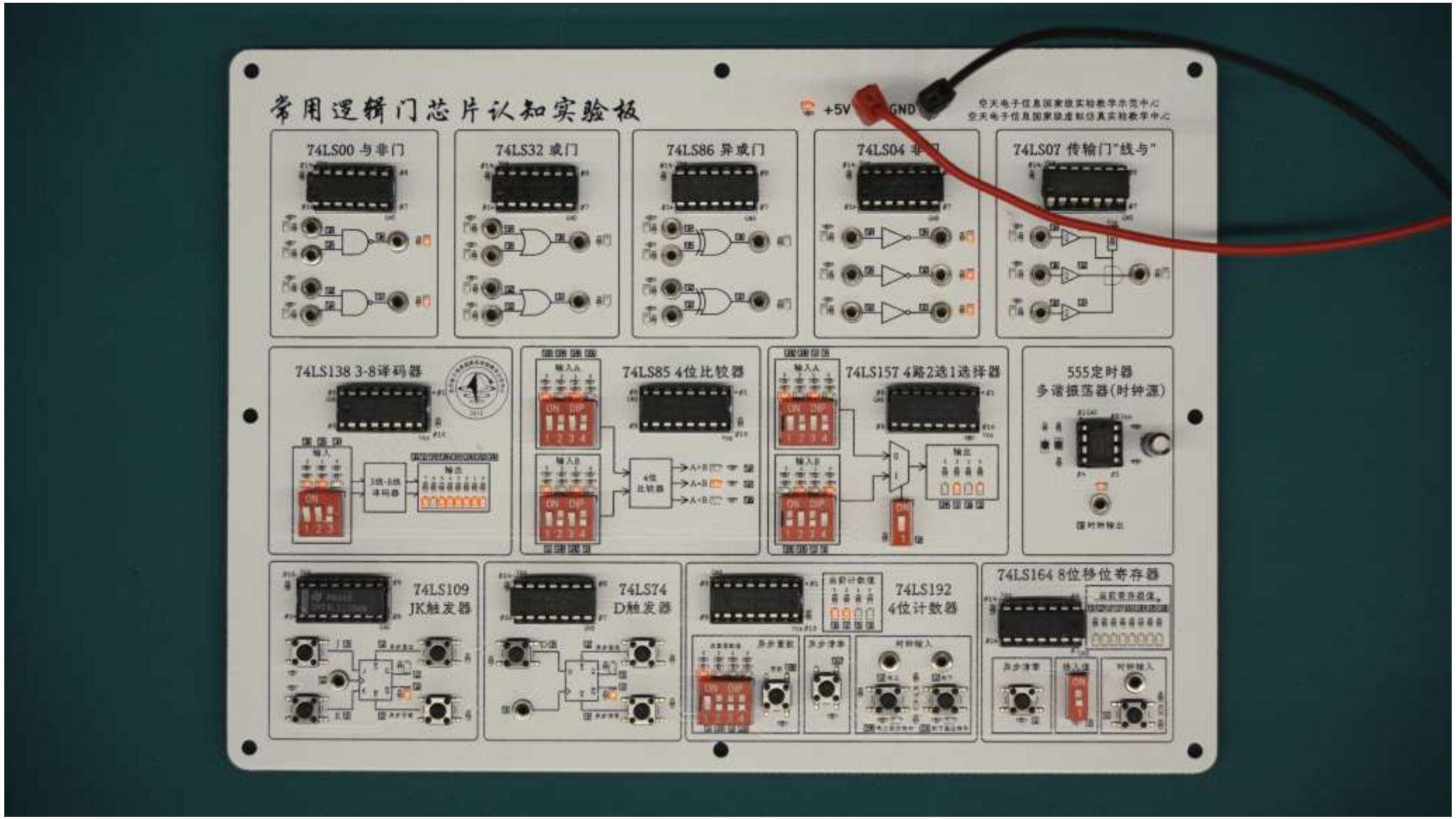
■ 实验步骤

- ➢ 实验板上电（5V供电）
- ➢ 按照右图所示将输出连接示波器
- ➢ 示波器的参考波形如下





演示视频


常用逻辑门芯片认知实验板
+5V
GND
74LS00 与非门
74LS32 或门
74LS86 异或门
74LS04 非门
74LS07 传输门"线与"
74LS138 3-8译码器
74LS85 4位比较器
74LS157 4路2选1选择器
555定时器
多谐振荡器(时钟源)
74LS109
JK触发器
74LS74
D触发器
74LS192
4位计数器
74LS164 8位移位寄存器

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